BundleFuse: A Technical Publication Argument

Definitions, exact identities, and the evidence supporting them

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Central claim. Historical item relations and evidence shared through item attributes provide complementary estimates of bundle compatibility. BundleFuse constructs and combines these estimates explicitly. Candidate-frequency backoff and multi-item terms are additional modeling choices whose benefits must be established empirically.

Scope. The equations below describe the binary-incidence, full-feature research formulation used for the four supplied POG/Spotify splits. They explain the common construction and its refinements; they are not an exhaustive specification of every modality or of the retail/commerce variants. Numerical findings in Section 6 use only the two completed POG mechanism studies available when this note was revised. The music mechanism studies are pending.

1 The prediction problem and the starting estimator

Let $\mathcal{I} = \{1, \dots, n\}$ be a fixed catalogue with $n > 1$ items. Let $S \subset \mathcal{I}$ be a nonempty observed set and let $c \in \mathcal{I} \setminus S$ be a candidate. The objective is to assign a real ranking score $\theta(c \mid S)$ to each candidate. A larger score indicates a preferred completion; the score is not defined as a probability. Hidden completion targets are available for validation and evaluation, but are not arguments of the scoring function.

Let m be the number of training bundles and $B \in \{0,1\}^{m \times n}$ their incidence matrix: $B_{ei} = 1$ precisely when training bundle e contains item i. Define

$$x_S(i) = \mathbf{1}[i \in S], \qquad f_i = \sum_{e=1}^{m} B_{ei}, \qquad d_e = \sum_{i=1}^{n} B_{ei}.$$

Here $\mathbf{1}[\cdot]$ is the indicator of a statement, f_i counts training appearances, and d_e is bundle size. An elementary score is

$$r_{\text{count}}(c \mid S) = (B^\top B x_S)_c = \sum_{s \in S} \sum_{e=1}^{m} B_{ec} B_{es}. \tag{1}$$

For $c \notin S$, this counts recorded co-memberships with the observed items. It motivates two modifications. First, observed counts can be dominated by frequent items and large bundles. Second, a zero count does not distinguish incompatibility from a lack of observations. These are reasons to consider modifications, not proofs that any particular modification improves ranking.

2 Construct historical estimates and extend their coverage

2.1 Weighted historical overlap

Define diagonal matrices $D \in \mathbb{R}^{n \times n}$ and $V \in \mathbb{R}^{m \times m}$ by

$$D_{ii} = (f_i + 5)^{-1/2}, \qquad V_{ee} = (d_e + 1)^{-1/2}.$$

The basic weighted overlap vector is

$$p(S) = B^\top VBDx_S. \tag{2}$$

For equal co-membership counts, common observed items and large training bundles contribute less. D acts on the observed-item side of this expression. Thus the operator in (2) is not generally symmetric. The constants 5, 1 and the exponents specify this version of the method; they are not derived optimal values.

For propagation, define $E_{ee} = (d_e + 1)^{-1}$ and

$$L = \mathrm{off}(DB^\top EBD), \tag{3}$$

where $\mathrm{off}(M)$ replaces a square matrix's diagonal by zero. The propagated score is $Lp(S)$. This uses indirect historical relationships. L is symmetric, but is not in general row-stochastic. Equation (2) itself retains its diagonal. Although that diagonal does not contribute directly to an eligible candidate's score, it affects observed-item entries used in propagation and standardization. It must not be silently removed from the definition of p.

Raw co-occurrence, positive pointwise mutual information, and collaborative singular value decomposition provide other summaries of historical relations. They reuse history and need not be statistically independent. Their inclusion is a choice of candidate estimators; their necessity is an empirical question.

2.2 Relations shared through categories and descriptions

Let $g_i \in \{1, \ldots, q\}$ be item i's category, and let $A \in \{0, 1\}^{n \times q}$ satisfy $A_{ia} = \mathbf{1}[g_i = a]$. Define $T \in \mathbb{R}^{q \times q}$ so that T_{ab} scores candidate category a against observed category b. A category-based score has the form

$$r_{\text{category}}(S) = ATA^\top x_S. \tag{4}$$

Pooling training observations across items can supply support when an exact pair in (1) has never occurred. Signed diagonal entries of T can encode additive penalties for repeated categories. The implemented category table and penalties are specified by the domain; (4) is their operator form, not a universal formula for estimating T. Such a relation need not be a similarity or a positive-semidefinite kernel.

Let $F \in \mathbb{R}^{n \times d}$ contain fixed item feature rows F_i. A descriptive score used in the full-ranking path is

$$r_F(S) = F\left(\frac{1}{|S|}\sum_{s \in S} F_s\right)^\top = \frac{1}{|S|}FF^\top x_S. \tag{5}$$

The feature dimension d and the row construction depend on the modality. Features can distinguish candidates absent from the training bundles. Resemblance is only one compatibility cue: distinct product categories may complement one another, whereas artist or album repetition may be useful in music.

For a supplied feature row X_i, let ℓ_i be its centering-group label. Fashion uses $\ell_i = g_i$; music uses labels obtained by clustering feature vectors, not artist labels. For each nonempty group a, define $\bar{X}_a = |\{i : \ell_i = a\}|^{-1} \sum_{i:\ell_i = a} X_i$. A residual feature starts from $X_i - \bar{X}_{\ell_i}$ and then normalizes each row. The residuals have zero group mean *before* row normalization. The normalized rows need not retain zero group mean, and this operation does not establish independence from category or group identity.

To state the feature bridge, define row normalization $\mathcal{N}(v) = v/(\|v\|_2 + 10^{-8})$, applied separately to each row of a matrix. For this construction, let F be the row-normalized concatenation of the

modality residuals in fashion, or the normalized audio residuals in music. Its graph-adjusted form is

$$\widetilde{F} = \mathcal{N}(F + \tfrac{1}{2}\mathcal{N}(LF)). \tag{6}$$

It mixes descriptions with descriptions propagated through historical relations. Category relations and this bridge therefore use both attributes and history. The two evidence families are functional groupings, not disjoint information sources.

3 Define the combined score before interpreting it

3.1 Standardization and the additive core

Let $\mathcal{H}$ be the available signal index set and let $r_h(S) \in \mathbb{R}^n$ denote raw signal $h \in \mathcal{H}$. Define

$$a_h(S) = \frac{1}{n}\sum_{i=1}^{n} r_h(i \mid S), \tag{7}$$

$$b_h(S) = \left[\frac{1}{n-1}\sum_{i=1}^{n}(r_h(i \mid S) - a_h(S))^2\right]^{1/2}, \tag{8}$$

$$z_h(c \mid S) = \mathcal{Z}_S[r_h]_c = \frac{r_h(c \mid S) - a_h(S)}{b_h(S) + \varepsilon}, \qquad \varepsilon = 10^{-9}. \tag{9}$$

The divisor $n-1$ specifies the Torch full-ranking path considered here. Some NumPy basket variants use divisor n and are outside this numerical specification. All catalogue items, including S, enter these statistics; items in S are excluded when recommendations are ranked. These operations depend on S and fixed model artifacts, not on the query's hidden targets. This establishes a dependency property of inference, not an audit of upstream feature training or dataset provenance.

Let $w_h \geq 0$ be a validation-selected weight. Without multi-item refinements, let z_h^0 denote each standardized signal and define

$$Y_0(c \mid S) = \sum_{h \in \mathcal{H}} w_h z_h^0(c \mid S). \tag{10}$$

Standardization makes the raw scales comparable; it does not make the signals independent or equally reliable. Because its mean and denominator depend on S, (10) need not equal a single fixed matrix times x_S.

3.2 Precisely defined multi-item refinements

We next define the optional corrections so that the eventual full score is explicit. For each training bundle, let $k_e(S) = \sum_{s \in S} B_{es}$ and $o_e(S) = V_{ee}\sum_{s \in S} B_{es}D_{ss}$. Define the incidence surplus

$$\delta_c(S) = \sum_{e=1}^{m} B_{ec}o_e(S)\,\beta\,[k_e(S) - 1]_+, \qquad \beta = 1.5, \qquad [x]_+ = \max(x, 0). \tag{11}$$

Only training bundles containing at least two observed items contribute. Replace p by $p + \delta$ and Lp by $Lp + L\delta$ in their respective raw channels, leaving other channels unchanged. Write z_h^1 for the separately standardized modified channels and $Y_1 = \sum_h w_h z_h^1$.

For the second refinement, let $W = \mathrm{off}(B^\top VBD)$ and let v_i be the selected normalized descriptor row for item i: the title residual in POG and the audio residual in Spotify. The category label g_i used below is distinct from the music centering-group label ℓ_i. Define candidate-to-seed support and seed similarity, respectively, by

$$q_{cs} = W_{cs} + 0.15\,[v_c \cdot v_s]_+, \qquad G_{st} = \tfrac{1}{2}\mathbf{1}[g_s = g_t] + \tfrac{1}{2}\,[v_s \cdot v_t]_+.$$

Set $K_{ss} = 0$. For distinct seeds s, t, use $K_{st} = [1 - G_{st}]_+$ for fashion and $K_{st} = 1 + G_{st}$ for music. The raw Rao-based score and its standardized version are

$$R(c \mid S) = \frac{\sum_{s,t \in S} q_{cs} K_{st} q_{ct}}{\sum_{s \in S} q_{cs} + 0.05}, \qquad z_R = \mathcal{Z}_S[R]. \tag{12}$$

This evaluates support involving pairs of observed items under the chosen category/ descriptor relationship. Both δ and R vanish when $|S| = 1$.

For selected $\alpha \in [0, 1]$ and $\gamma \geq 0$, define the full combined estimate

$$Y = (1 - \alpha)Y_0 + \alpha Y_1 + \gamma z_R. \tag{13}$$

The mixture is formed *after* each raw channel is standardized. In general, $\mathcal{Z}_S[p + \alpha\delta] \neq (1 - \alpha)\mathcal{Z}_S[p] + \alpha\mathcal{Z}_S[p + \delta]$. These corrections expand the construction; they are not assumed to improve every dataset.

3.3 The restricted fallback and its relation to the full estimate

Let $\mathcal{Q} \subset \mathcal{H}$ be the available fallback-eligible channels: category, soft-category, text-style, image-style and bridge for fashion; category, group homophily, audio-style and bridge for music. Let $\mathcal{A} = \{h : w_h > 0\}$. Define

$$\mathcal{P} = \begin{cases} \mathcal{Q} \cap \mathcal{A}, & \mathcal{Q} \cap \mathcal{A} \neq \varnothing, \\ \mathcal{Q}, & \text{otherwise.} \end{cases}$$

Assume $\mathcal{Q}$ is nonempty, as in these supplied datasets. For $h \in \mathcal{P}$, let $\widetilde{w}_h = w_h$ if its positive weight was retained, and let $\widetilde{w}_h = 0.5$ otherwise. Let z_{pop} denote the standardized $\log(1 + f_i)$ score, with weight $w_{\text{pop}} = 0$ if it was not selected. Then

$$P = \sum_{h \in \mathcal{P}} \widetilde{w}_h z_h^0 + \gamma z_R + w_{\text{pop}} z_{\text{pop}}. \tag{14}$$

The channels in $\mathcal{Q}$ are unaffected by the incidence modification, so $z_h^0 = z_h^1$ there. The fallback retains selected popularity and Rao terms as well as category/descriptive channels. It is neither independent of history nor disjoint from Y. Its restricted channel set is a modeling choice motivated by coverage, not a theorem requiring this exact set.

4 Frequency-dependent backoff: an exact interpretation

Let $s_0 \geq 0$ be a validation-selected fallback scale and set $\mu_0 = s_0 P$. Let $\kappa \geq 0$ be the selected backoff strength. Define the final score by

$$\theta(c \mid S) = \begin{cases} Y(c \mid S), & \kappa = 0, \\ \dfrac{f_c Y(c \mid S) + \kappa \mu_0(c \mid S)}{f_c + \kappa}, & \kappa > 0. \end{cases} \tag{15}$$

When enabled, backoff increases the fallback's relative weight for candidates with less training history. Treating frequency as a measure of the useful amount of history is an assumption; frequency is not a calibrated estimate of uncertainty.

Proposition 1 (Pointwise regularization identity)**.** *Fix S, c, Y, μ_0 and $\kappa > 0$. Over $t \in \mathbb{R}$, define*

$$J_c(t) = f_c(t - Y)^2 + \kappa(t - \mu_0)^2.$$

The score in (15) is the unique minimizer of J_c.

Proof. $J_c'(t) = 2f_c(t - Y) + 2\kappa(t - \mu_0)$ and $J_c''(t) = 2(f_c + \kappa) > 0$. Setting the derivative to zero gives $t = (f_c Y + \kappa\mu_0)/(f_c + \kappa)$. □

This identity explains how a fixed pair of scores is combined. It is not the objective optimized by the entire fitting pipeline, and does not prove optimality of the frequency weight, fallback construction, or ranking risk. If $f_c = \kappa = 0$, the quadratic is constant; returning Y is the explicit disabled-backoff convention.

A Gaussian posterior interpretation requires additional distributional assumptions. The archived four supplied-split fits have raw slopes approximately $-0.273, -2.528, -5.336, -5.869$ when mean squared disagreement between Y and the scaled fallback is regressed on $1/f_c$ over warm items. Those signs do not support the posited positive inverse-frequency variance relation in those fits. They are not formal rejection tests, and disagreement between two scores is not ground-truth prediction error. The procedure clips variance estimates and selects backoff settings using validation. Thus the exact identity above is a firmer explanation than a claim of empirically validated posterior uncertainty.

5 Consequences of shared terms

For $\kappa > 0$, define $\rho_c = f_c/(f_c + \kappa)$ and $H = Y - P$. Substitution gives

$$\theta = \rho_c H + [\rho_c + (1 - \rho_c)s_0]P. \tag{16}$$

At $s_0 = 1$, this reduces to $P + \rho_c H$. At a different scale, the coefficient of P also varies by candidate. H is defined algebraically; it can be interpreted as the remaining selected channels when shared coefficients agree, as in the four archived selections. Default fallback weights require separate interpretation.

For comparison, let $a \in [0, 1]$ be a candidate-independent fallback fraction. Its score is

$$\theta_a = (1 - a)Y + as_0 P = (1 - a)H + (1 - a + as_0)P.$$

When Y and P use a common fixed signal dictionary, this constant mixture is a reweighting of that dictionary. Improvements over a coarse, sequentially selected weight vector do not by themselves establish a new representational capability. Frequency-dependent backoff permits different coefficients for different candidates; its practical advantage over constant reweighting must be measured.

Proposition 2 (Ordering within the cold subset)**.** *Let $\mathcal{C}_0 = \{c \in \mathcal{I} \setminus S : f_c = 0\}$. Hold all component scores and weights fixed. Suppose $Y(c \mid S) - P(c \mid S) = b(S)$ for all $c \in \mathcal{C}_0$, where $b(S)$ does not depend on c. If $s_0 > 0$ and $\kappa > 0$, enabling backoff preserves the ordering of candidates within $\mathcal{C}_0$, including ties.*

Proof. For $c, d \in \mathcal{C}_0$, the disabled score difference is $Y(c \mid S) - Y(d \mid S) = P(c \mid S) - P(d \mid S)$. The enabled difference is $s_0[P(c \mid S) - P(d \mid S)]$, which has the same sign. □

This is a statement in exact real arithmetic. With exact scores, an identical item-ID tie rule preserves the within-subset ranking. Floating-point evaluation can create or break near-ties; bit-identical numerical rankings do not follow from the proof. The proposition does not preserve rankings between cold and warm candidates, nor apply to a comparison that also retunes components. One sufficient condition is that $Y - P$ contains only standardized history channels whose raw scores are zero for every cold candidate. Catalogue standardization then contributes the same constant for each cold candidate within each such channel. Auxiliary histories, different fallback coefficients, or $s_0 = 0$ can invalidate the premises.

6 Fitting and the empirical argument

Weights are selected by a finite greedy coordinate search on validation ranking quality, followed by optional-term and backoff selection. This is a fitted procedure with order dependence and finite search resolution; it is not guaranteed to find a global optimum. In the mechanism study, every reported recipe is frozen before test scoring. Catalogue features are available transductively. No conclusion here asserts that external feature encoders were trained without evaluation-item exposure.

The completed studies yield the following Recall@20 values:

Construction	POG	POG_dense
Historical family	0.030875	0.375379
Shared category/descriptive family	0.044542	0.112917
Combined, without refinements	0.051458	0.390928
Full BundleFuse	0.054458	0.395223

The historical family includes popularity. The family-only models are separately validation-selected; they are not frozen-weight removals. The combined model exceeds both families individually in Recall@20 and NDCG@20 under the current correction reserving all 96 planned comparisons. This supports complementary predictive information in the tested constructions. It does not prove every channel necessary or attribute the contrast between datasets solely to density.

With matched fallback scale, frequency backoff exceeds the selected constant mixture on POG: Recall@20 is 0.054458 versus 0.051833, with corrected evidence in both metrics. Under unit scale, that superiority is not established. On POG_dense, the corresponding matched-scale recalls are 0.392893 and 0.394353; their difference is not established. Absence of a significant difference is not an equivalence result.

On POG, cold-target recall in full-catalogue retrieval increases from 0.039884 without backoff to 0.044069 with matched-scale frequency backoff, while within-cold Recall@20 remains 0.075222. These are descriptive bucket diagnostics, conditional on queries containing cold targets. They agree with the ordering proposition's predicted distinction. The evaluated core variants on POG_dense retrieve no cold targets in the global top 20, despite nonzero within-cold recall; universal cold-start success is therefore unsupported.

The five tested selection conditions give Recall@20 ranges $[0.050958, 0.054458]$ on POG and $[0.385736, 0.395896]$ on POG_dense. These descriptive ranges show that selection sensitivity is material. The best variant must not be chosen using test results. The studies are follow-up analyses on previously inspected benchmarks, with uncertainty conditional on fitted models and fixed splits. The final joint correction across datasets awaits the music runs.

7 The resulting publication claim

The supported central argument is that explicit historical estimates and evidence shared through item attributes improve one another in the completed experiments. Backoff and multi-item corrections extend that construction, with benefits that depend on the evaluated conditions. The contribution must concern the particular score construction and its demonstrated behavior relative to appropriate controls. It cannot rest solely on combining information sources, avoiding a neural encoder, or rewriting a convex mixture as a quadratic minimizer.

This positioning is consistent with prior work: EASE establishes strong explicit item-relation modeling [1]; CLHE and RaMen already combine several information sources for bundle construc-

tion [2, 3]; empirical-Bayes product search already combines behavioral estimates with descriptive priors [4]. The complete related-work and benchmark-provenance comparison remains necessary.

Audit trail. The equations were checked against the scoring and fusion implementation at commit `80ff927`. The new POG results come from the dataset-level `summary.json` files under `results/publication_mechanisms`; the four historical variance fits come from the supplied-split selections at `4516eab`. Historical and current experiment records are distinguished. An independent synthetic-catalogue audit checked 52 equation cases against the implementation, including fallback-default weights, propagation, incidence, Rao scores, and enabled/disabled backoff. The largest absolute discrepancy was 4.77×10^{-7}, consistent with the tested float32 computations. Numerical agreement checks implementation correspondence, not statistical optimality. The audit is retained in `results/argument_audit/equations.json`. The method source and running experiments were not changed by this document revision.

References

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